Macroeconomics PhD Intro
Lecture 2: The Solow Model

Dynamics, Growth, Convergence, and Propagation

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Learning Goals

- Build the Solow model from assumptions to dynamics.
- Derive steady states, balanced growth, and convergence speed.
- Connect theory to facts on growth, returns, and factor shares.
- Use the model for impulse responses and log-linear analysis.
- Understand how the Solow framework serves as the core of virtually all modern business-cycle models.

Motivation: The Puzzle of the Stable Capital-Output Ratio

One of the central long-run economic facts is the stability of the capital-output ratio K_t/Y_t over time: capital is roughly **three times annual GDP** in most advanced economies.

Earlier growth theories (Harrod-Domar) took this stability as a *technological* assumption—they simply built it into the model. But capital is only one production input. Production can be organized in many ways: highly capital-intensive or highly labor-intensive. Given the many possibilities, it is **not obvious** why K_t/Y_t should be nearly constant at the macro level.

Solow's Starting Point

Reconcile the stable capital-output ratio with the *substitutability* of capital and labor. This reconciliation gave rise to:

1. A framework for analyzing macroeconomic dynamics.
2. Systematic methods for measuring technological change (growth accounting).

What the Solow Model Will Deliver

The aggregate production function has three variables that can affect GDP growth: technology A_t, capital K_t, and labor L_t. The Solow model focuses on the **endogenous accumulation of capital** K_t.

We will see that K_t reacts not only to the saving rate s but also to A_t and L_t. After solving the model (which delivers a stable K/Y ratio), we focus on two main takeaways:

1. The fundamental source of long-run growth in per capita income is the growth in A_t.
2. If all parameter values are common across economies, different countries converge to the same income per capita—both in level and growth rate—in the long run.

Road Map

Basic model $\rightarrow$ steady state and dynamics $\rightarrow$ growing economy $\rightarrow$ prices and shares $\rightarrow$ convergence speed $\rightarrow$ business cycles and IRFs $\rightarrow$ log-linearization.

Stylized Facts to Keep in Mind

- **Fact 1:** The capital-output ratio K_t/Y_t is roughly stable over long horizons (approximately 3 in the U.S.).
- **Fact 2:** GDP per capita grows persistently at a roughly constant rate in advanced economies.
- **Fact 3:** The return to physical capital has been nearly constant over time.
- **Fact 4:** Factor shares (capital share $\approx 1/3$, labor share $\approx 2/3$) are roughly stable in the long run.
- **Fact 5:** There is conditional convergence across similar economies, but not strong unconditional convergence across all countries.

The Solow model will account for all five of these facts with a single, parsimonious framework.

Technology and Assumptions

We start with the simplest version: no technological progress, no population growth.

Aggregate Production Function

$$Y_t = F(K_t, L_t)$$

We interpret Y_t as GDP. The function $F(K, L)$ satisfies:

1. $F(K, L)$ is strictly increasing in both K and L.
2. $F(K, L)$ is strictly quasiconcave (strictly convex isoquants).
3. Constant returns to scale: $F(cK, cL) = cF(K, L)$ for any $c > 0$.
4. $F(0, L) = 0$: no output without capital.
5. Inada conditions: $\lim_{K \to 0} F_1(K, L) = \infty$ and $\lim_{K \to \infty} F_1(K, L) = 0$.

We also assume F is twice continuously differentiable, so that FOCs generate continuous functions.

Why These Assumptions Matter

- **CRS** allows reduction to a per-worker (or per-efficiency-worker) system—we collapse an n-dimensional problem into a one-dimensional difference equation.
- **Diminishing marginal product of capital** (from strict quasiconcavity) generates mean reversion of k_t: rich economies grow slower, poor economies grow faster.
- **Inada conditions** guarantee an interior steady state. They are stronger than strictly necessary—it suffices that $\lim_{k\to 0} f'(k) > \delta/s$ and $\lim_{k\to\infty} f'(k) < \delta/s$—but they simplify the exposition.
- **Strict quasiconcavity** gives well-behaved substitution between factors: isoquants are smooth and convex.

Economic Logic

At low k, the marginal product of capital is high, so investment return is high and capital grows. At high k, the marginal product is low and depreciation pressure dominates, so capital shrinks. The economy gravitates toward an intermediate steady state.

What Does "Neoclassical" Mean Here?

Neoclassical Production Side

- Smooth substitution between capital and labor (convex isoquants).
- Diminishing marginal products for each input.
- Constant returns to scale at the aggregate level.
- Competitive pricing: factors paid their marginal products.

- **Dynamic implication:** Strong convergence force through diminishing returns—the economy always returns to its steady state.
- **Empirical implication:** Disciplined mapping from factor shares and growth facts to structural parameters (α, δ, etc.).
- **Contrast with Harrod-Domar:** Fixed proportions (Leontief) technology implied knife-edge instability. The neoclassical substitutability of inputs removes this instability entirely.

Per-Capita Representation

In the basic model, population and hours per capita are both constant and normalized to 1. Therefore:

- Define per-worker (= per-capita) variables: $y_t \equiv Y_t/L_t$, $k_t \equiv K_t/L_t$.
- By CRS: $Y_t/L_t = F(K_t/L_t, 1)$.
- Define $f(k) \equiv F(k, 1)$. Then:

$$y_t = f(k_t) \tag{3.1}$$

The intensive-form production function $f(\cdot)$ inherits all the properties of F: it is strictly increasing, strictly concave, and satisfies the Inada conditions in k.

Key point: Because of the normalization $L_t = 1$, per-capita and aggregate variables are the same in the basic model. The distinction will matter once we introduce population growth. Appendix

Capital Accumulation and Goods Market

The second centerpiece of the Solow model is the equation describing the evolution of capital:

Capital Accumulation

$$k_{t+1} = i_t + (1 - \delta)k_t \tag{3.2}$$

where i_t is investment and $\delta \in (0, 1)$ is capital's depreciation rate. The existing capital stock loses value δk_t while being used.

Goods Market Clearing

$$y_t = c_t + i_t \tag{3.3}$$

Output is either consumed or invested. We implicitly assume goods are homogeneous—they can be used for both consumption and investment. Because output equals total income, total saving must equal total investment. (In an open economy with trade, this need not hold.)

Saving Rule and Fundamental Equation

Instead of explicitly modeling consumption-saving decisions, the Solow model assumes consumers mechanically save a constant fraction of income:

Behavioral Closure

$$i_t = sy_t, \quad s \in (0, 1) \tag{3.4}$$

This is the simplifying assumption that will be relaxed in Chapter 4, where saving arises from optimizing behavior.

Inserting (3.4) into (3.2) and using (3.1):

Fundamental Solow Equation

$$k_{t+1} = (1 - \delta)k_t + sf(k_t) \tag{3.5}$$

This difference equation is the **heart of the Solow model**. It expresses the entire dynamics of capital as a function of one endogenous variable k_t and three exogenous objects: δ, s, and f.

Appendix

Economic Meaning of the Fundamental Equation

$$k_{t+1} = \underbrace{(1-\delta)k_t}_{\text{surviving capital}} + \underbrace{sf(k_t)}_{\text{new investment}}$$

- The only endogenous variable on the RHS is k_t. Given k_0, the entire future path $\{k_{t+1}\}_{t=0}^{\infty}$ is determined.
- Once we have the capital path, all other aggregates follow immediately:

Immediate Implications

$$y_t = f(k_t), \qquad i_t = sf(k_t), \qquad c_t = (1-s)f(k_t)$$

Transitional dynamics of output, investment, and consumption are all inherited from the path of k_t. This is the power of the one-dimensional state representation.

Steady State Definition

A **steady state** is a situation where k_t is constant over time: $k_t = \bar{k}$ for all t. From (3.5):

Steady-State Condition

$$\bar{k} = (1-\delta)\bar{k} + sf(\bar{k}) \quad \Longrightarrow \quad \delta\bar{k} = sf(\bar{k})$$

- In steady state, investment exactly replaces depreciation: $i = \delta\bar{k}$.
- All variables are constant: $\bar{y} = f(\bar{k})$, $\bar{c} = (1-s)f(\bar{k})$, $\bar{i} = sf(\bar{k})$.
- No growth in the basic model—the economy simply sits at $\bar{k}$ forever.

Existence and Uniqueness of the Steady State

To see that a unique positive $\bar{k}$ exists, plot δk and $sf(k)$ against k:

- δk is a straight line through the origin with slope δ.
- $sf(k)$ is strictly increasing and strictly concave, reflecting diminishing returns.
- Near $k = 0$: the Inada condition $f'(0) = \infty$ implies $sf(k)$ has infinite slope, so it starts above δk.
- As $k \to \infty$: the Inada condition $f'(\infty) = 0$ implies $sf(k)$ eventually grows slower than δk.
- These two curves must cross exactly once at a positive $\bar{k}$.

Note: The Inada conditions are stronger than necessary. It suffices that $\lim_{k\to 0} f'(k) > \delta/s$ (existence) and $\lim_{k\to\infty} f'(k) < \delta/s$ (uniqueness). Appendix

Cobb-Douglas Closed Form

A particularly useful production function is Cobb-Douglas: $F(K, L) = K^{\alpha} L^{1-\alpha}$, so $f(k) = k^{\alpha}$ with $\alpha \in (0, 1)$. This satisfies all our assumptions, including Inada conditions.

Analytical Steady State

From $\delta \bar{k} = s \bar{k}^{\alpha}$:

$$\bar{k} = \left(\frac{s}{\delta}\right)^{\frac{1}{1-\alpha}} \tag{3.6}$$

- $\bar{k}$ is increasing in s: higher saving $\Rightarrow$ more investment $\Rightarrow$ larger capital stock.
- $\bar{k}$ is decreasing in δ: faster depreciation $\Rightarrow$ more capital lost each period $\Rightarrow$ lower steady state.
- Because $\bar{y} = f(\bar{k}) = \bar{k}^{\alpha}$, steady-state output is also increasing in s and decreasing in δ.

Appendix

Steady-State Capital-Output Ratio

Going back to our motivating fact: in the Cobb-Douglas case, the steady-state k/y ratio is:

Key Ratio

$$\frac{\bar{k}}{\bar{y}} = \frac{s}{\delta}$$

- In the basic model, k_t/y_t is trivially constant in steady state (all variables are constant).
- But as we will see, even in a *growing* economy where both k_t and y_t increase over time, this ratio eventually stabilizes—matching the data.
- $\bar{k}/\bar{y}$ is larger when s is larger (more saving builds up capital relative to output).
- $\bar{k}/\bar{y}$ is smaller when δ is larger (faster depreciation erodes the stock).
- With $s \approx 0.2$ and $\delta \approx 0.05$, we get $\bar{k}/\bar{y} \approx 4$, broadly consistent with U.S. data.

Appendix

Comparative Statics in Elasticity Form

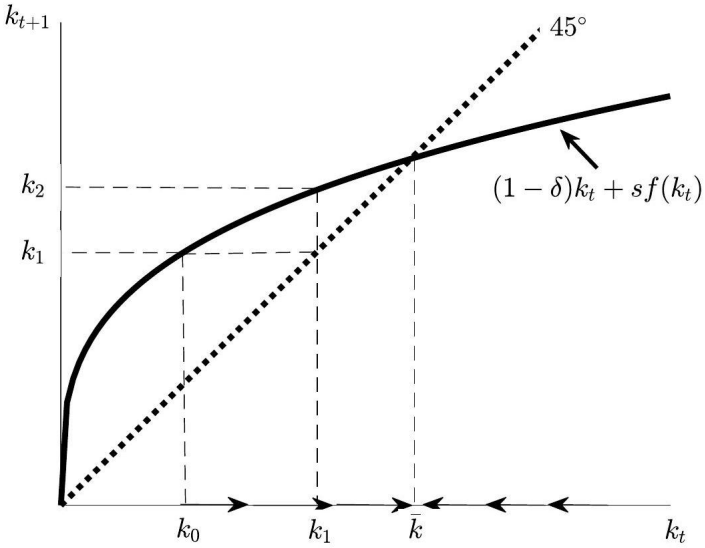
From $\bar{k} = (s/\delta)^{1/(1-\alpha)}$

$$\frac{\partial \ln \bar{k}}{\partial \ln s} = \frac{1}{1-\alpha}, \qquad \frac{\partial \ln \bar{k}}{\partial \ln \delta} = -\frac{1}{1-\alpha}$$

- With $\alpha = 1/3$: a 1% increase in s raises $\bar{k}$ by 1.5%.
- Output level response: $\partial \ln \bar{y} / \partial \ln s = \alpha/(1-\alpha)$.
- With $\alpha = 1/3$: a 1% increase in s raises $\bar{y}$ by only 0.5%—diminishing returns dampen the output response.
- Higher curvature (lower α) weakens sensitivity: with strong diminishing returns, the extra capital produces little extra output.

Appendix

Global Dynamics: Cobweb Diagram



Proof Sketch of Monotone Convergence

Suppose $k_t < \bar{k}$. We show:

1. $k_{t+1} > k_t$: Because $k_t < \bar{k}$, we have $sf(k_t) > \delta k_t$ (the investment curve is above the depreciation line), so $k_{t+1} - k_t = sf(k_t) - \delta k_t > 0$.
2. $k_{t+1} < \bar{k}$: Because $k_t < \bar{k}$ and the RHS of (3.5) is increasing in k_t, the value at k_t is less than the value at $\bar{k}$, which equals $\bar{k}$.

Therefore $k_t < k_{t+1} < \bar{k}$. Repeating: the sequence $\{k_t\}$ is monotone increasing and bounded above by $\bar{k}$.

By the **Monotone Convergence Theorem**, the sequence has a limit. The limit must satisfy the steady-state equation, and since the steady state is unique, the limit is $\bar{k}$.

The symmetric argument for $k_t > \bar{k}$ gives a monotone decreasing sequence bounded below by $\bar{k}$.

Appendix

Convergence Intuition: Two Opposing Forces

Rewrite the capital accumulation equation as:

$$k_{t+1} - k_t = \underbrace{sf(k_t)}_{\text{investment force}} - \underbrace{\delta k_t}_{\text{depreciation force}}$$

- When k_t is **small**: output per unit of capital $f(k_t)/k_t$ is large (diminishing returns). The constant saving rate implies large gross investment relative to the capital stock. Investment force wins.
- When k_t is **large**: output per unit of capital is small. Gross investment cannot cover total depreciation δk_t. Depreciation force wins.

In growth-rate form:

Growth Rate of Capital

$$\frac{k_{t+1} - k_t}{k_t} = \frac{sf(k_t)}{k_t} - \delta$$

Since $f''(k) < 0$ and $f(0) = 0$, the average product $f(k)/k$ is decreasing in k. This generates the negative relationship between the growth rate and the level—the engine of convergence.

Cobweb Method: General Setup

One-Dimensional Map

$$k_{t+1} = G(k_t), \quad G(k) \equiv (1-\delta)k + sf(k)$$

- Steady states are fixed points: $k^* = G(k^*)$, i.e., where the map crosses the 45-degree line.
- **Cobweb iteration:** Given k_t, move vertically to $G(\cdot)$ to read off k_{t+1}, then move horizontally to the 45-degree line to place k_{t+1} on the horizontal axis. Repeat.
- The geometry around intersections determines local dynamics: the slope $G'(k^*)$ tells us everything about local behavior.
- This technique generalizes to any one-dimensional dynamical system—not just Solow.

Local Cobweb Stability: Derivation

Define the deviation $x_t \equiv k_t - k^*$. First-order Taylor expansion around k^*:

$$G(k_t) \approx G(k^*) + G'(k^*)(k_t - k^*)$$

Since $G(k^*) = k^*$ (fixed point), we get:

$$x_{t+1} \approx G'(k^*)\, x_t$$

This is a simple linear recursion. Its solution is $x_t = [G'(k^*)]^t\, x_0$, so:

- If $|G'(k^*)| < 1$: deviations shrink $\Rightarrow$ **locally stable**.
- If $|G'(k^*)| > 1$: deviations grow $\Rightarrow$ **locally unstable**.
- If $|G'(k^*)| = 1$: borderline case, higher-order terms matter.

The *sign* of $G'(k^*)$ determines whether convergence is monotone or oscillatory.

Appendix

Cobweb Case 1: Monotone Convergence ($0 < G'(k^*) < 1$)

Condition

$$0 < G'(k^*) < 1$$

- Deviations keep the same sign and shrink each period: $x_t \to 0$ monotonically.
- Trajectory approaches k^* from one side without overshooting.
- The **baseline neoclassical Solow model** is always in this case near steady state, because $G'(\bar{k}) = 1 - \delta + sf'(\bar{k}) = 1 - \delta(1 - \alpha) \in (0, 1)$ for standard parameter values.

Cobweb Case 2: Oscillatory Convergence ($-1 < G'(k^*) < 0$)

Condition

$$-1 < G'(k^*) < 0$$

- The sign of x_t flips each period: the path overshoots the steady state repeatedly.
- But the absolute deviation shrinks, so oscillations are damped.
- Requires a locally *decreasing* map around the fixed point—unusual for the standard Solow model, but can arise with non-standard production functions.
- Think of the bakery metaphor: too many ovens can reduce output if they crowd out workers and dough space.

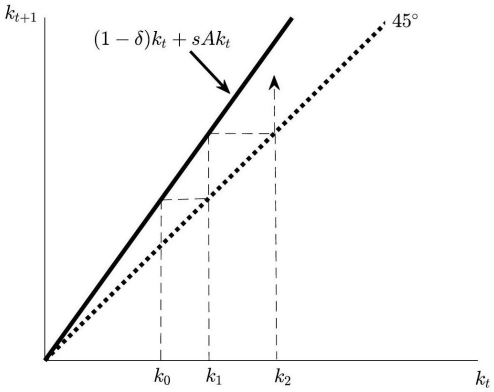
Cobweb Case 3: Instability ($|G'(k^*)| > 1$)

Conditions

$$G'(k^*) > 1 \quad \text{or} \quad G'(k^*) < -1$$

- Deviations expand over time: local divergence from the fixed point.
- If $G'(k^*) > 1$: monotone explosion away from steady state.
- If $G'(k^*) < -1$: paths alternate in sign and explode—or settle into cycles or chaotic dynamics (forever non-monotone behavior that never settles to a repeated pattern).
- This is where multiple steady states and nonlinear attractors can emerge globally.

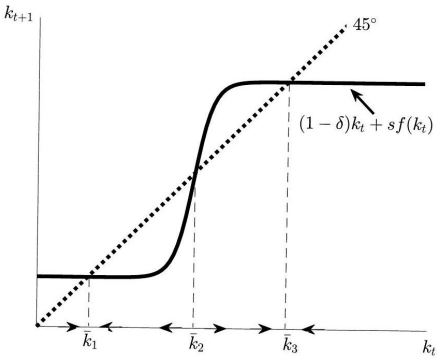
Endogenous Growth Case



Consider the case where $F_1(k, 1)$ is uniformly above δ/s. No positive steady state exists—k_t grows without bound.

- Special case: $y_t = Ak_t$ (linear in capital, $\alpha = 1$ in Cobb-Douglas). Known as the "AK model."
- Growth is "endogenous": it comes from within, not from exogenous technology.
- Two identical countries starting with different k_0 remain forever different—the gap in percentage terms stays constant.
- Since labor commands $\approx 2/3$ of income, the AK setup is empirically implausible as a standalone model, but the concept is developed further in Chapter 13 with multiple accumulable inputs.

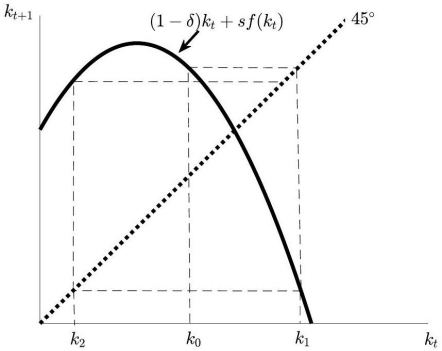
Poverty Traps



If $f(k)$ is not globally concave—e.g., there is a middle region with increasing returns (large infrastructure investments, building of transportation networks)—then the curve $k_{t+1} = G(k_t)$ can cross the 45-degree line multiple times.

- In Figure 3.3: three steady states $\bar{k}_1 < \bar{k}_2 < \bar{k}_3$.
- $\bar{k}_1$ and $\bar{k}_3$ are **stable**; $\bar{k}_2$ is **unstable**.
- A country starting with $k_0 < \bar{k}_2$ converges to $\bar{k}_1$—the **poverty trap**.
- To escape: push k_t above $\bar{k}_2$, e.g., by temporarily raising s high enough that $\bar{k}_1$ and $\bar{k}_2$ disappear as intersections.

Complex Dynamics and Chaos



If $f(k)$ actually *declines* at very high k (more ovens than floor space in the bakery), then $G(k)$ can become decreasing. In this case:

- Convergence, if it occurs at all, will not be monotone.
- k_t can exhibit forever oscillating or even **chaotic** dynamics: great sensitivity to initial conditions, behavior that never settles to a repeated pattern.
- Mathematically: chaos requires $|G'(k^*)| > 1$ with a non-monotone map.
- More esoteric in macroeconomics, but illustrates the full range of dynamics a one-dimensional system can produce.

Policy Interpretation of Non-Convex Cases

- **Multiple steady states** imply history dependence and threshold effects. Initial conditions determine long-run outcomes.
- A **temporary policy** that lifts k_t above the unstable threshold can permanently alter the economy's trajectory—a "big push" argument for development policy.
- In practice, identifying true macro-level non-convexities (increasing returns at the aggregate level) has proved empirically difficult. The growth literature has not produced robust evidence for the kind of increasing returns needed.

Caution

These cases are useful for understanding possible mechanisms, but baseline neoclassical concavity remains the empirical benchmark.

Extending the Model: Why Growth?

In the basic model, the long-run outcome is a steady state with **zero growth**. This is clearly counterfactual—GDP per capita grows persistently in the data.

To address the growth facts from Chapter 2, we extend the model by allowing:

1. The labor input L_t to grow at rate n (population growth and changes in hours worked).
2. The technology level A_t to grow at rate γ (labor-augmenting technical change).

Harrod-Neutral Technical Change

The extended production function takes the form:

$$Y_t = F(K_t, A_t L_t)$$

- $A_t L_t$ is the total number of **efficiency units of labor** (or effective labor).
- Technology A_t improves the labor input: this is **labor-augmenting** (Harrod-neutral) technical change.
- **Uzawa's Theorem (1961):** Labor-augmenting technical change is the *only* form of technical progress consistent with exact balanced growth (where aggregate variables grow at constant rates). This is a deep theoretical result, not just a convenient assumption.
- Growth rate of A_t: γ. Growth rate of L_t: n.
- Note: n has two origins—growing population and changes in hours worked per person (which has been declining in the long run).

Transform to Efficiency Units

The same algebra as in the basic model yields $K_{t+1} = (1-\delta)K_t + sF(K_t, A_t L_t)$.

Define Capital per Efficiency Unit

$$\tilde{k}_t \equiv \frac{K_t}{A_t L_t}$$

Dividing both sides by $A_t L_t$ and noting that $K_{t+1}/(A_t L_t) = (1+\gamma)(1+n)\tilde{k}_{t+1}$:

Fundamental Equation with Growth

$$(1+\gamma)(1+n)\tilde{k}_{t+1} = (1-\delta)\tilde{k}_t + sf(\tilde{k}_t) \tag{3.7}$$

After normalization by $A_t L_t$, we recover a difference equation very similar to (3.5). The factor $(1+\gamma)(1+n)$ on the LHS reflects the extra "depreciation" from growing efficiency units: maintaining a given $\tilde{k}$ is harder when A and L grow.

Appendix

Balanced Growth Path

The concept corresponding to the steady state is the **balanced growth path** (BGP). Along the BGP, $\tilde{k}_t$ is constant and all aggregate variables grow at constant rates.

Setting $\tilde{k}_{t+1} = \tilde{k}_t = \overline{\tilde{k}}$ in (3.7):

$$[(1+\gamma)(1+n) - (1-\delta)]\overline{\tilde{k}} = sf(\overline{\tilde{k}})$$

With Cobb-Douglas $f(\tilde{k}) = \tilde{k}^{\alpha}$:

BGP Capital per Efficiency Unit

$$\overline{\tilde{k}} = \left(\frac{s}{(1+\gamma)(1+n) + \delta - 1}\right)^{\frac{1}{1-\alpha}} \tag{3.8}$$

Notice that γ and n enter like additional depreciation: faster growth in A and L means each unit of untransformed capital must "make up for" more effective depreciation.

Appendix

Long-Run Growth Rate of Income per Capita

Income per capita is:

$$y_t = \frac{Y_t}{L_t} = f(\tilde{k}_t) \cdot A_t$$

Along the BGP, $f(\tilde{k}_t) = f(\overline{\tilde{k}})$ is constant. Therefore:

$$\frac{y_{t+1} - y_t}{y_t} = \frac{A_{t+1} - A_t}{A_t} = \gamma$$

Fundamental Result

The growth rate of per capita income in the long run equals γ—the rate of technological progress. **No other parameter affects it.** Encouraging saving (raising s) does *not* affect the long-run growth rate.

This is perhaps the most surprising and important prediction of the Solow model: the engine of sustained growth is exogenous technology, not capital accumulation.

Appendix

Transition Dynamics and Convergence

In the short run, $\tilde{k}_t$ changes over time, affecting growth:

$$\frac{y_{t+1} - y_t}{y_t} = \frac{f(\tilde{k}_{t+1})}{f(\tilde{k}_t)}(1+\gamma) - 1$$

- If $\tilde{k}_t < \overline{\tilde{k}}$: $\tilde{k}$ is growing, so $f(\tilde{k}_{t+1})/f(\tilde{k}_t) > 1$ and per-capita growth **exceeds** γ. The economy is "catching up."
- If $\tilde{k}_t > \overline{\tilde{k}}$: $\tilde{k}$ is falling, so per-capita growth is **below** γ. The economy is "winding down" to its BGP.

Convergence Prediction

Economies below their own BGP grow faster; economies above it grow slower. Poor countries (relative to their own BGP) grow faster during transition—this is the convergence prediction of the Solow model.

Note: this is *conditional* convergence—it requires countries to share the same parameter values.

Level Effect vs Growth Effect of Saving

What happens when s permanently increases?

- $\overline{\tilde{k}}$ rises (from Eq. 3.8), so the **level** of y_t on the BGP is permanently higher.
- During the transition to the new BGP, per-capita growth temporarily exceeds γ as $\tilde{k}_t$ rises toward its new, higher value.
- But once the economy reaches the new BGP, per-capita growth returns to γ.

Core Distinction

The Solow model gives strong **level effects** of saving, but **no permanent growth effect**. Only changes in γ can permanently alter long-run growth.

Quantitative example: With our calibration, doubling s from 0.1 to 0.2 raises normalized output by about 17%. The transition takes decades, temporarily boosting growth, but eventually the economy returns to growing at rate γ.

Competitive Firm and Return to Capital

To connect the model to data on returns and factor shares, we need prices. Suppose firms maximize profit under competitive markets:

Firm Problem

$$\max_{K_t, A_t L_t} F(K_t, A_t L_t) - r_t K_t - w_t A_t L_t \tag{3.9}$$

Here, output is the numéraire (price = 1). Capital is owned by consumers and rented to firms at rate r_t; w_t is the wage per efficiency unit of labor.

From the FOC for K_t:

$$r_t = F_1(K_t, A_t L_t) = f'(\tilde{k}_t)$$

Along the BGP, $\tilde{k}_t$ is constant, so r_t is constant—matching the data fact of a roughly stable return to capital.

Appendix

Labor Share and Capital Share

The wage per efficiency unit satisfies: $w_t = F_2(K_t, A_t L_t)$.

When F has constant returns to scale, it is homogeneous of degree one. By **Euler's Theorem** for homogeneous functions:

$$Y_t = K_t F_1(K_t, A_t L_t) + A_t L_t F_2(K_t, A_t L_t)$$

Dividing by Y_t:

$$1 = \underbrace{\frac{r_t K_t}{Y_t}}_{\text{capital share}} + \underbrace{\frac{w_t A_t L_t}{Y_t}}_{\text{labor share}}$$

Capital share plus labor share equals one. Along the BGP, since both r_t and K_t/Y_t are constant, the capital share is constant. The labor share, being one minus the capital share, is also constant—matching the data.

Appendix

Cobb-Douglas Benchmark for Factor Shares

If $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$

- Capital share $= r_t K_t / Y_t = \alpha$.
- Labor share $= w_t A_t L_t / Y_t = 1 - \alpha$.
- These are constant **even away from the BGP**—at any point on the transition path.

This is a special property of Cobb-Douglas. For more general CES production functions $Y = [\alpha K^{\rho} + (1-\alpha)(AL)^{\rho}]^{1/\rho}$, factor shares vary with $K/(AL)$ unless $\rho = 0$ (Cobb-Douglas).

Key insight: The Cobb-Douglas form is very convenient because it often simplifies algebra and leads to clean expressions. But this simplicity can be deceiving: it makes fundamental forces going in opposite directions cancel, rendering them invisible.

Why Share Stability Matters Quantitatively

- Factor share data disciplines α directly: from U.S. national income accounts, $\alpha \approx 1/3$.
- With Cobb-Douglas, stable shares become a built-in structural feature, not an additional implication to check.
- The value of α pins down convergence speed λ and all transition dynamics.
- The substitution elasticity between inputs is exactly 1 for Cobb-Douglas. Empirical estimates at the aggregate level rarely suggest major departures from 1, supporting the functional form choice.

Empirical Caveat

If measured shares trend strongly (as some recent work on declining labor shares suggests), Cobb-Douglas may require augmentation: human capital, markups, or a non-unit elasticity of substitution.

Local Linearization: No-Growth Case

To measure how fast the economy converges, linearize the fundamental equation around $\bar{k}$:

$$\Delta k_{t+1} = \left(1 - \delta + sf'(\bar{k})\right) \Delta k_t \tag{3.10}$$

where $\Delta k_t \equiv k_t - \bar{k}$. Define the **convergence speed**:

$$\lambda \equiv \delta - sf'(\bar{k}), \quad \text{so that} \quad \frac{k_{t+1} - \bar{k}}{\bar{k}} = (1 - \lambda)\frac{k_t - \bar{k}}{\bar{k}}$$

A larger λ means faster convergence (the gap shrinks more each period).

For Cobb-Douglas, using the steady-state condition:

$$\lambda = \delta(1 - \alpha) \tag{3.11}$$

Convergence is faster when α is small (stronger diminishing returns) and δ is large. Note: s does *not* affect λ in the Cobb-Douglas case—two opposing forces cancel exactly. Appendix

Why s Doesn't Affect λ in Cobb-Douglas

This deserves explanation because it is a subtle cancellation:

1. For a given k_t, a larger s means more investment, so k_t has a *larger* impact on k_{t+1}. This would *slow* convergence (the map is steeper).
2. But a larger s also raises the steady-state $\bar{k}$, and at higher $\bar{k}$ the marginal product $f'(\bar{k})$ is smaller (diminishing returns). This makes the map *flatter* near steady state, speeding up convergence.

With Cobb-Douglas, these two forces exactly offset. For non-Cobb-Douglas production functions, s generally does affect λ.

Lesson: Cobb-Douglas simplicity can hide important economic forces. Always check whether a result is general or an artifact of the functional form.

Local Linearization: Growth Case

With population and technology growth, linearizing (3.7) around $\overline{\tilde{k}}$:

$$(1+\gamma)(1+n)\Delta\tilde{k}_{t+1} = \left(1-\delta+sf'(\overline{\tilde{k}})\right)\Delta\tilde{k}_t \tag{3.12}$$

The convergence speed becomes:

$$\lambda = 1 - \frac{1-\delta+sf'(\overline{\tilde{k}})}{(1+\gamma)(1+n)}$$

For Cobb-Douglas:

$$\lambda = (1-\alpha)\left(1-\frac{1-\delta}{(1+\gamma)(1+n)}\right) \tag{3.13}$$

Population growth and technology growth both increase λ (speed up convergence), because they act like additional effective depreciation.

Appendix

Speed Intuition

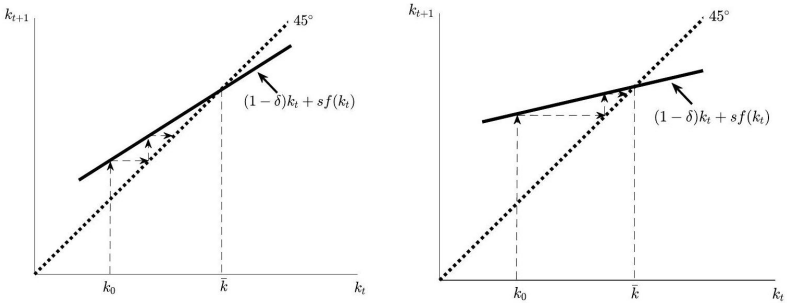


Figure 3.5: A higher λ means a flatter $G(\cdot)$ curve near the steady state, which geometrically means fewer cobweb steps to reach $\bar{k}$—faster convergence.

From λ to Half-Life

The local convergence law is: $\text{gap}_{t+1} = (1-\lambda)\cdot\text{gap}_t$. After h periods:

$$\text{gap}_h = (1-\lambda)^h \cdot \text{gap}_0$$

The **half-life** h solves $(1-\lambda)^h = 1/2$:

$$h = \frac{\ln(1/2)}{\ln(1-\lambda)} \approx \frac{0.693}{\lambda}$$

- With $\lambda = 0.049$ (model prediction): $h \approx 14$ years.
- With $\lambda = 0.02$ (empirical estimate): $h \approx 35$ years.
- The model predicts faster convergence than we observe—a key tension we will return to.

Convergence in Data: All Countries vs. OECD

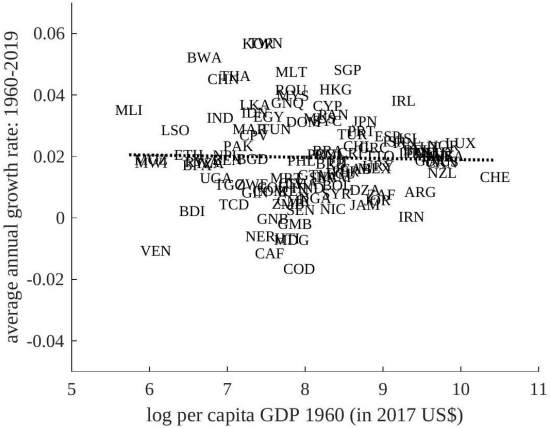


Figure 3.6: All countries, 1960–2019. Source: PWT 10.0.

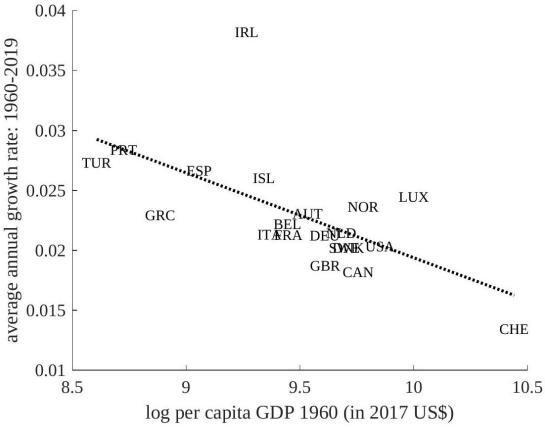


Figure 3.7: OECD countries, 1960–2019. Source: PWT 10.0.

- **All countries:** No systematic tendency for poor countries to grow faster—no unconditional convergence. But this does *not* reject the Solow model, which predicts *conditional*

Recent Convergence Patterns and Calibration

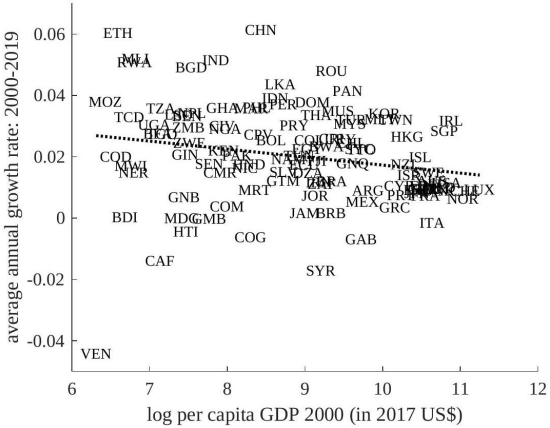


Figure 3.8: All countries, 2000–2019. Source: PWT 10.0.

- Kremer, Willis, and You (2022) show that unconditional convergence has emerged in recent decades.
- Their argument: policies, institutions, and human capital have become more similar across countries, making underlying parameters more homogeneous.

Standard calibration: $\alpha = 1/3$, $\gamma = 0.02$, $n = 0.01$, $\delta \approx 0.046$. This gives $\lambda = 0.049$, but the empirical estimate is $\lambda \approx 0.015$–0.03 (Barro and Sala-i-Martin, 2004). The model over-predicts

Solow as Propagation Core

The Solow model's usefulness goes far beyond growth theory. Due to its ability to account for the broad features of macroeconomic data, it constitutes the **core of virtually all modern business-cycle models**.

- Business-cycle theories analyze how the economy reacts to shocks—the "propagation mechanism."
- Many cycle models are essentially Solow plus shocks: the state equation remains a modified version of (3.5).
- Three main types of shocks have been studied in the literature:
  1. Neutral technology shocks (RBC tradition).
  2. Investment-specific technology shocks.
  3. Demand disturbances (Keynesian tradition).

Shock 1: Neutral Technology (RBC Model)

Production function: $y_t = A_t F(k_t, \ell_t)$ with variable labor ℓ_t.

- Shock: A_t switches between high and low values. Economy moves toward different steady states.
- With constant s and ℓ, the model cannot match key business-cycle facts: (i) ℓ_t comoves with the cycle, (ii) i_t is more volatile than y_t, (iii) c_t is less volatile than y_t.
- So we allow s and ℓ to react to (k_t, A_t):

$$k_{t+1} = (1-\delta)k_t + s(k_t, A_t)\, A_t\, F(k_t, \ell(k_t, A_t))$$

Note: Under Cobb-Douglas, the distinction between Harrod-neutral ($F(K, AL)$) and Hicks-neutral ($AF(K, L)$) is inessential: $K^\alpha (AL)^{1-\alpha} = A^{1-\alpha} K^\alpha L^{1-\alpha}$.

Shock 2: Investment-Specific Technology

Goods market clearing becomes $y_t = c_t + i_t/\nu_t$, where ν_t captures the relative efficiency of producing investment goods:

- When ν_t is high, it is cheaper to produce investment goods—one unit of consumption goods yields ν_t units of investment.
- This reflects a two-sector structure: the investment-goods sector has its own productivity.

$$k_{t+1} = s\nu_t F(k_t, \ell) + (1 - \delta)k_t$$

Shock 3: Demand-Driven Disturbance

A Keynesian-flavored model: c_t is exogenous and moves over time (the demand shock).

- Total demand: $c_t + i_t = c_t/(1-s)$ (since $i_t/c_t = s/(1-s)$).
- When demand falls short of full-capacity output $F(k_t, \ell)$, total output is demand-determined and a fraction u_t of the labor force becomes unemployed.
- $u_t = u(c_t, k_t)$ from: $F(k_t, \ell(1-u_t)) = c_t/(1-s)$.

$$k_{t+1} = sF(k_t, \ell(1-u(c_t, k_t))) + (1-\delta)k_t$$

This framework is Keynesian in spirit but begs the question of *why* output can be below full capacity. Understanding the frictions behind demand shortfalls is central to later chapters.

Impulse-Response Setup

In all three shock models, k_{t+1} is a function of k_t and a shock. To visualize the dynamics, we compute **impulse-response functions** (IRFs).

Setup: Use the RBC version with Cobb-Douglas, exogenous s, and fixed ℓ:

$$k_{t+1} = sA_t k_t^{\alpha} \ell^{1-\alpha} + (1-\delta)k_t \tag{3.14}$$

- Before period 0: $A_t = \bar{A}$, k_t is near $\bar{k}$.
- At $t = 0$: $A_0 = (1+\varepsilon)\bar{A}$ (1% shock).
- For $t \geq 1$: $A_t = (1+\rho^t\varepsilon)\bar{A}$, with $\rho = 0.9$ (persistence).
- Starting from $k_0 = \bar{k}$, simulate (3.14) forward.

Appendix

Impulse Response in the Solow-RBC Setup

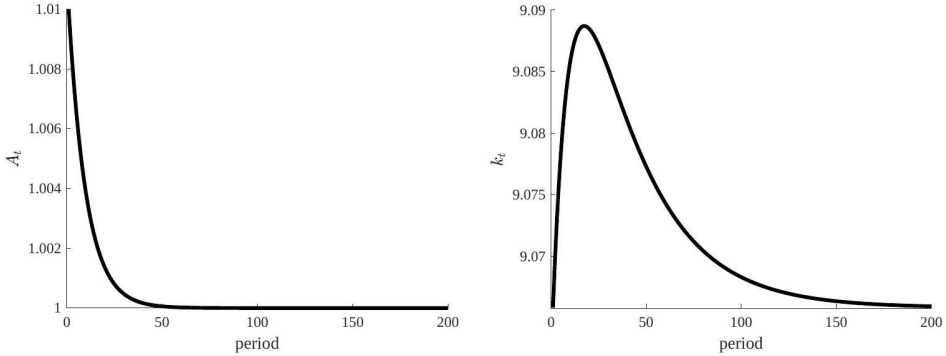


Figure 3.9: Left: A_t mean-reverts quickly. Right: k_t responds gradually and persistently.

- k_t rises from $\bar{k} = 9.066$ to a peak of 9.089 (+0.25%), then gradually returns—a hump-shaped response.
- While A_t's deviation fades by period ∼50, capital's response persists much longer.
- The peak capital deviation (0.25%) is much smaller than the initial shock (1%)—capital acts as a buffer.

Why Log-Linearization?

- Converts nonlinear dynamics into tractable **linear recursions** that can be solved analytically.
- Variables are expressed as **percent deviations** from steady state—natural economic units.
- Usually accurate for moderate shocks near the BGP. In our calibrated Solow model, the maximum approximation error is about 0.00001 in units of K_t—indistinguishable from the exact solution when plotted.
- This technique is the workhorse of modern quantitative macroeconomics: DSGE models, RBC analysis, and New Keynesian models all rely heavily on log-linearization.

Log-Deviation Definitions

An arbitrary variable x_t is expressed as:

$$x_t = \bar{x} e^{\hat{x}_t}, \qquad \hat{x}_t \equiv \log\left(\frac{x_t}{\bar{x}}\right) \tag{3.15}$$

Since $\log(x_t/\bar{x}) \approx (x_t - \bar{x})/\bar{x}$ for small deviations (first-order Taylor expansion):

$$\hat{x}_t \approx \frac{x_t - \bar{x}}{\bar{x}} \quad \text{(percent deviation from steady state)}$$

And the exponential approximation:

$$\bar{x} e^{\hat{x}_t} \approx \bar{x}(1 + \hat{x}_t) \tag{3.16}$$

These two approximations—replacing $e^{\hat{x}}$ with $1 + \hat{x}$ and $\log(1 + z)$ with z—are the building blocks of log-linearization.

Linearized Capital Dynamics

Apply (3.15)–(3.16) to the Solow-RBC law (3.14):

$$\bar{k}e^{\hat{k}_{t+1}} = s\bar{A}e^{\hat{A}_t}(\bar{k}e^{\hat{k}_t})^{\alpha}\ell^{1-\alpha} + (1-\delta)\bar{k}e^{\hat{k}_t}$$

First-order approximation around zero deviations, using the steady-state identity $\delta\bar{k} = s\bar{A}\bar{k}^{\alpha}\ell^{1-\alpha}$:

Log-Linearized Capital Law

$$\hat{k}_{t+1} = (1-\delta(1-\alpha))\hat{k}_t + \delta\hat{A}_t \tag{3.18}$$

- Let $\lambda \equiv \delta(1-\alpha)$—the same convergence speed as in (3.11).
- The log-linearization procedure is essentially the same as the linearization for convergence speed: both apply first-order Taylor approximation to (3.14).
- Capital responds to shocks with coefficient δ and has persistence $(1-\lambda)$.

Appendix

Closed-Form Response and Output Deviation

With $\hat{A}_0 = \varepsilon$ and $\hat{A}_t = \rho^t \varepsilon$ for $t \geq 1$, solving (3.18) forward:

$$\hat{k}_{t+1} = \sum_{\tau=0}^{t} \rho^{t-\tau}(1-\lambda)^{\tau}\delta\varepsilon = \rho^t \frac{1 - \left(\frac{1-\lambda}{\rho}\right)^{t+1}}{1 - \frac{1-\lambda}{\rho}} \delta\varepsilon$$

Output deviation (from log-linearizing $y_t = A_t k_t^{\alpha} \ell^{1-\alpha}$):

$$\hat{y}_t = \hat{A}_t + \alpha \hat{k}_t$$

Output responds through two channels: the **direct effect** of the technology shock ($\hat{A}_t$) and the **propagated effect** through accumulated capital ($\alpha \hat{k}_t$). Log-linearization gives us these closed-form expressions—impossible to obtain from the nonlinear system.

Appendix

Reading the IRF Formula

- The weights in the sum combine **shock persistence** ρ and **capital persistence** $(1-\lambda)$.
- If $\rho > (1-\lambda)$: shock persistence dominates early dynamics. Capital builds up as long as the technology shock is active.
- If $\rho < (1-\lambda)$: capital's own persistence dominates. The hump in capital outlasts the shock itself.
- Larger λ (faster convergence) shortens capital memory and compresses the hump-shaped response.
- Output has a direct term ($\hat{A}_t$, which decays as ρ^t) plus a propagated term ($\alpha\hat{k}_t$, which decays more slowly). This two-component structure is what generates the characteristic hump shape of output IRFs.

Level-Linearization Alternative

When the log formulation is inconvenient, we can linearize in levels:

$$\Delta k_{t+1} = g_k \Delta k_t + g_A \Delta A_t$$

where $g(k_t, A_t) \equiv (1-\delta)k_t + sA_t f(k_t)$, and g_k, g_A are partial derivatives evaluated at steady state.

For Cobb-Douglas:

- $g_k = 1 - \delta(1-\alpha)$ (same persistence as in log-linear version).
- $g_A = s^{1/(1-\alpha)}\delta^{-\alpha/(1-\alpha)}\bar{A}^{\alpha/(1-\alpha)}$.

The level-linearization and log-linearization give equivalent results for small deviations. The choice between them is one of convenience: log-deviations give percentage interpretations directly, while level-deviations may be easier when some variables can take negative values.

What the Solow Model Delivers

1. A disciplined dynamic core linking capital accumulation to growth, convergence, and the stability of key macroeconomic ratios.
2. A clean mapping from parameters (α, s, δ, γ, n) to steady states, transition speeds, and long-run growth rates.
3. Matches five stylized facts: stable K/Y, persistent per-capita growth, stable returns to capital, stable factor shares, and conditional convergence.
4. A foundation for business-cycle propagation: virtually all modern macro models build on a version of the Solow framework.
5. Analytical tractability via log-linearization, enabling closed-form impulse responses and quantitative analysis.

What the Solow Model Leaves Exogenous

- **Saving behavior:** The saving rate s is assumed constant and exogenous. In reality, saving responds to interest rates, wealth, expectations about future income, etc.
- **Technology process:** The growth rate γ of A_t is taken as given. What drives technological progress? R&D investment, institutions, human capital?
- **Labor supply:** In the basic model, labor is fixed. In reality, hours worked respond to wages, taxes, and the business cycle.

Next Lecture (Chapter 4)

Endogenous saving via **dynamic optimization**. Cass (1965) and Koopmans (1963) add microeconomic foundations: saving arises from rational, forward-looking consumers maximizing lifetime utility. This gives the same long-run implications as Solow, but allows for richer policy analysis and normative evaluation.

A1. Aggregate to Per-Capita Form

1. Start: $Y_t = F(K_t, L_t)$.
2. Define $y_t \equiv Y_t/L_t$, $k_t \equiv K_t/L_t$.
3. CRS implies $Y_t/L_t = F(K_t/L_t, 1)$: divide both arguments by L_t.
4. Define $f(k) \equiv F(k, 1)$, so $y_t = f(k_t)$.

Properties of f: Since F is strictly increasing in K, so is f in k. Since F is CRS, $F_{11} < 0$ (given strict quasiconcavity), so $f'' < 0$. The Inada conditions transfer: $f'(0) = \infty$, $f'(\infty) = 0$. Back

A2. Deriving the Fundamental Equation

1. Capital law: $k_{t+1} = i_t + (1-\delta)k_t$.
2. Saving rule: $i_t = sy_t$.
3. Production: $y_t = f(k_t)$.
4. Substitute (2) and (3) into (1): $k_{t+1} = (1-\delta)k_t + sf(k_t)$.

This is a first-order nonlinear difference equation in k_t. Given $k_0 > 0$, it generates the entire time path of the economy.

Back

A3. Why Unique Positive Steady State?

Solve $\delta k = sf(k)$, equivalently $\delta = sf(k)/k = s \cdot \text{APK}(k)$.

- LHS: constant $\delta > 0$.
- RHS: $s \cdot f(k)/k$. By Inada, $f'(0) = \infty \Rightarrow f(k)/k \to \infty$ as $k \to 0$.
- By concavity, $f(k)/k$ is strictly decreasing in k (since $f'' < 0$ and $f(0) = 0$).
- By Inada, $f'(\infty) = 0 \Rightarrow f(k)/k \to 0$ as $k \to \infty$.
- A continuous strictly decreasing function from ∞ to 0 crosses any positive horizontal line exactly once.

Back

A4. Derivation of Eq. (3.6)

With $f(k) = k^\alpha$:

$$\delta \bar{k} = s\bar{k}^\alpha \quad \Rightarrow \quad \bar{k}^{1-\alpha} = \frac{s}{\delta} \quad \Rightarrow \quad \bar{k} = \left(\frac{s}{\delta}\right)^{\frac{1}{1-\alpha}}$$

Steady-state output: $\bar{y} = \bar{k}^\alpha = (s/\delta)^{\alpha/(1-\alpha)}$.
Steady-state consumption: $\bar{c} = (1-s)\bar{y} = (1-s)(s/\delta)^{\alpha/(1-\alpha)}$.

Back

A5. Monotone Convergence Detail

Case $k_t < \bar{k}$:

- $G(k_t) > k_t$ because $sf(k_t) > \delta k_t$ when $k < \bar{k}$. So $k_{t+1} > k_t$.
- $G(k_t) < G(\bar{k}) = \bar{k}$ because G is increasing and $k_t < \bar{k}$. So $k_{t+1} < \bar{k}$.
- Combined: $k_t < k_{t+1} < \bar{k}$.

Case $k_t > \bar{k}$: Symmetric argument gives $\bar{k} < k_{t+1} < k_t$.

In both cases, the sequence is monotone and bounded. By the Monotone Convergence Theorem, it has a limit k^*. Taking limits in $k_{t+1} = G(k_t)$ gives $k^* = G(k^*)$, and by uniqueness of the steady state, $k^* = \bar{k}$.

Back

A6. Derivation of $\bar{k}/\bar{y} = s/\delta$

From the steady-state condition $\delta\bar{k} = sf(\bar{k}) = s\bar{y}$:

$$\frac{\bar{k}}{\bar{y}} = \frac{s}{\delta}$$

This holds for *any* production function satisfying our assumptions, not just Cobb-Douglas. The capital-output ratio in steady state depends only on the saving rate and the depreciation rate. Back

A7. Derivation of Eq. (3.7)

Start from $K_{t+1} = (1-\delta)K_t + sF(K_t, A_tL_t)$. Divide both sides by A_tL_t:

$$\frac{K_{t+1}}{A_tL_t} = (1-\delta)\frac{K_t}{A_tL_t} + s\frac{F(K_t, A_tL_t)}{A_tL_t}$$

The LHS is:

$$\frac{K_{t+1}}{A_tL_t} = \frac{A_{t+1}}{A_t} \cdot \frac{L_{t+1}}{L_t} \cdot \frac{K_{t+1}}{A_{t+1}L_{t+1}} = (1+\gamma)(1+n)\tilde{k}_{t+1}$$

By CRS, $F(K_t, A_tL_t)/(A_tL_t) = F(K_t/(A_tL_t), 1) = f(\tilde{k}_t)$. Therefore:

$$(1+\gamma)(1+n)\tilde{k}_{t+1} = (1-\delta)\tilde{k}_t + sf(\tilde{k}_t)$$

Back

A8. Derivation of Eq. (3.8)

Set $\tilde{k}_{t+1} = \tilde{k}_t = \overline{\tilde{k}}$ in (3.7):

$$(1+\gamma)(1+n)\overline{\tilde{k}} = (1-\delta)\overline{\tilde{k}} + s\overline{\tilde{k}}^{\alpha}$$

$$[(1+\gamma)(1+n) - (1-\delta)]\overline{\tilde{k}} = s\overline{\tilde{k}}^{\alpha}$$

$$\overline{\tilde{k}}^{1-\alpha} = \frac{s}{(1+\gamma)(1+n)+\delta-1}$$

$$\overline{\tilde{k}} = \left(\frac{s}{(1+\gamma)(1+n)+\delta-1}\right)^{\frac{1}{1-\alpha}}$$

Back

A9. Why Long-Run Per-Capita Growth Is γ

$$y_t = \frac{Y_t}{L_t} = \frac{F(K_t, A_t L_t)}{L_t} = A_t \cdot \frac{F(K_t, A_t L_t)}{A_t L_t} = A_t \cdot f(\tilde{k}_t)$$

On BGP, $\tilde{k}_t = \overline{\tilde{k}}$ is constant, so $f(\tilde{k}_t)$ is constant. Therefore:

$$\frac{y_{t+1} - y_t}{y_t} = \frac{f(\overline{\tilde{k}})A_{t+1} - f(\overline{\tilde{k}})A_t}{f(\overline{\tilde{k}})A_t} = \frac{A_{t+1} - A_t}{A_t} = \gamma$$

The $f(\overline{\tilde{k}})$ terms cancel, leaving only the growth rate of technology.

Back

A10. Deriving $r_t = f'(\tilde{k}_t)$

- FOC of firm problem: $r_t = F_1(K_t, A_t L_t)$.
- From CRS: $f(\tilde{k}_t) \equiv F(\tilde{k}_t, 1) = F(K_t, A_t L_t)/(A_t L_t)$.
- Differentiate both sides with respect to K_t:

$$f'(\tilde{k}_t) \cdot \frac{1}{A_t L_t} = \frac{F_1(K_t, A_t L_t)}{A_t L_t}$$

- Therefore $F_1(K_t, A_t L_t) = f'(\tilde{k}_t)$, i.e., $r_t = f'(\tilde{k}_t)$.

Back

A11. Factor Shares Under CRS

By Euler's theorem for homogeneous-of-degree-one functions:

$$Y_t = K_t F_1(K_t, A_t L_t) + A_t L_t F_2(K_t, A_t L_t)$$

Divide by Y_t:

$$1 = \frac{r_t K_t}{Y_t} + \frac{w_t A_t L_t}{Y_t}$$

Hence labor share $= 1-$ capital share. Both are constant along BGP since r_t and $K_t/Y_t = \tilde{k}_t/f(\tilde{k}_t)$ are both constant when $\tilde{k}_t = \overline{\tilde{k}}$.

Alternative: $w_t = f(\tilde{k}_t) - \tilde{k}_t f'(\tilde{k}_t)$, which is constant when $\tilde{k}_t$ is constant.

Back

A12. Deriving Eqs. (3.10)–(3.11)

Linearize $g(k) = (1-\delta)k + sf(k)$ at $\bar{k}$:

$$\Delta k_{t+1} = g'(\bar{k})\Delta k_t = (1 - \delta + sf'(\bar{k}))\Delta k_t$$

Define $\lambda = \delta - sf'(\bar{k})$, so $g'(\bar{k}) = 1 - \lambda$.

For Cobb-Douglas: $f'(k) = \alpha k^{\alpha-1}$. At steady state, $s\bar{k}^{\alpha-1} = \delta/\alpha \cdot \alpha = \delta$ (from $s\bar{k}^{\alpha} = \delta\bar{k}$, so $sf'(\bar{k}) = s\alpha\bar{k}^{\alpha-1} = \alpha\delta$). Therefore:

$$\lambda = \delta - \alpha\delta = \delta(1-\alpha)$$

Back

A13. Deriving Eqs. (3.12)–(3.13)

Linearize the growth-case map $(1+\gamma)(1+n)\tilde{k}_{t+1} = (1-\delta)\tilde{k}_t + sf(\tilde{k}_t)$ around $\overline{\tilde{k}}$:

$$(1+\gamma)(1+n)\Delta\tilde{k}_{t+1} = (1-\delta+sf'(\overline{\tilde{k}}))\Delta\tilde{k}_t$$

The convergence speed:

$$\lambda = 1 - \frac{1-\delta+sf'(\overline{\tilde{k}})}{(1+\gamma)(1+n)}$$

For Cobb-Douglas, $sf'(\overline{\tilde{k}}) = \alpha[(1+\gamma)(1+n)-1+\delta]$ (from the BGP condition). Substituting:

$$\lambda = (1-\alpha)\left(1 - \frac{1-\delta}{(1+\gamma)(1+n)}\right)$$

Back

A14. IRF Calibration Inputs

In the chapter illustration:

- $s = 0.2$, $\delta = 0.046$, $\alpha = 1/3$.
- $\ell = 1$, $\bar{A} = 1$.
- Initial shock $\varepsilon = 0.01$ (1%), persistence $\rho = 0.9$.
- Simulate Eq. (3.14) from $k_0 = \bar{k} = (s\bar{A}/\delta)^{1/(1-\alpha)} = (0.2/0.046)^{1.5} \approx 9.066$.
- $A_t = (1 + \rho^t \varepsilon)\bar{A}$ for $t = 0, 1, 2, \ldots$

The implied convergence speed is $\lambda = 0.046 \times 2/3 \approx 0.031$.

Back

A15. Derivation of Eq. (3.18)

Start from the transformed law:

$$\bar{k}e^{\hat{k}_{t+1}} = s\bar{A}e^{\hat{A}_t}(\bar{k}e^{\hat{k}_t})^{\alpha}\ell^{1-\alpha} + (1-\delta)\bar{k}e^{\hat{k}_t}$$

Apply $e^{\hat{x}} \approx 1 + \hat{x}$ and $(\bar{k}e^{\hat{k}_t})^{\alpha} \approx \bar{k}^{\alpha}(1 + \alpha\hat{k}_t)$:

$$\bar{k}(1 + \hat{k}_{t+1}) = s\bar{A}\bar{k}^{\alpha}\ell^{1-\alpha}(1 + \hat{A}_t)(1 + \alpha\hat{k}_t) + (1-\delta)\bar{k}(1 + \hat{k}_t)$$

Drop second-order terms ($\hat{A}_t \cdot \hat{k}_t \approx 0$) and use $\delta\bar{k} = s\bar{A}\bar{k}^{\alpha}\ell^{1-\alpha}$:

$$\bar{k}\hat{k}_{t+1} = \delta\bar{k}(\hat{A}_t + \alpha\hat{k}_t) + (1-\delta)\bar{k}\hat{k}_t$$

Divide by $\bar{k}$:

$$\hat{k}_{t+1} = (1 - \delta(1-\alpha))\hat{k}_t + \delta\hat{A}_t$$

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A16. Closed-Form Responses

Given $\hat{A}_t = \rho^t \varepsilon$ and Eq. (3.18), iterate forward from $\hat{k}_0 = 0$:

$$\begin{aligned}
\hat{k}_1 &= \delta\varepsilon \\
\hat{k}_2 &= (1-\lambda)\delta\varepsilon + \delta\rho\varepsilon \\
&\vdots \\
\hat{k}_{t+1} &= \sum_{\tau=0}^{t} \rho^{t-\tau}(1-\lambda)^{\tau}\delta\varepsilon = \rho^t \frac{1-\left(\frac{1-\lambda}{\rho}\right)^{t+1}}{1-\frac{1-\lambda}{\rho}}\delta\varepsilon
\end{aligned}$$

Output deviation: $\hat{y}_t = \hat{A}_t + \alpha\hat{k}_t = \rho^t\varepsilon + \alpha\hat{k}_t$.
With $\lambda = \delta(1-\alpha)$.

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A17. Cobweb Stability in Solow Form

For Solow, $G(k) = (1-\delta)k + sf(k)$, so:

$$G'(k) = 1 - \delta + sf'(k)$$

At k^*, local convergence requires $|G'(k^*)| < 1$, i.e.:

$$|1 - \delta + sf'(k^*)| < 1$$

Since $1 - \delta + sf'(k^*) > 0$ for reasonable parameters, this simplifies to:

$$1 - \delta + sf'(k^*) < 1 \quad \Leftrightarrow \quad sf'(k^*) < \delta \quad \Leftrightarrow \quad \lambda = \delta - sf'(k^*) > 0$$

In Cobb-Douglas: $\lambda = \delta(1-\alpha)$, so $0 < \lambda < 1$ for $\alpha \in (0,1)$ and any $\delta \in (0,1)$. Monotone convergence is guaranteed.

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A18. What Is an Elasticity?

General Definition

The **elasticity** of y with respect to x measures the percent change in y per percent change in x:

$$\varepsilon_{y,x} \equiv \frac{\partial y/y}{\partial x/x} = \frac{\partial y}{\partial x} \cdot \frac{x}{y} = \frac{\partial \ln y}{\partial \ln x}$$

Why use elasticities instead of plain derivatives?

- Elasticities are **unit-free**: they do not depend on whether capital is measured in dollars, euros, or physical units. A derivative $\partial \bar{k}/\partial s$ has units of "capital per unit of saving rate," which is hard to interpret. An elasticity says: "a 1% rise in s raises $\bar{k}$ by ε%."
- Elasticities give **proportional** comparisons: they tell you how large the response is *relative to the starting point*, which is economically more meaningful than the absolute change.
- The $\partial \ln y/\partial \ln x$ representation is especially convenient because many economic relationships become linear in logs (e.g., Cobb-Douglas: $\ln y = \alpha \ln k$).

A18 (cont.). Elasticity Applied to the Solow Steady State

Start from $\bar{k} = (s/\delta)^{1/(1-\alpha)}$. Take logs:

$$\ln \bar{k} = \frac{1}{1-\alpha}(\ln s - \ln \delta)$$

Differentiate with respect to $\ln s$ (holding δ fixed):

$$\frac{\partial \ln \bar{k}}{\partial \ln s} = \frac{1}{1-\alpha}$$

Interpretation: A 1% increase in the saving rate raises the steady-state capital stock by $1/(1-\alpha)$ percent.

For output, $\ln \bar{y} = \alpha \ln \bar{k}$, so:

$$\frac{\partial \ln \bar{y}}{\partial \ln s} = \alpha \cdot \frac{1}{1-\alpha} = \frac{\alpha}{1-\alpha}$$

With $\alpha = 1/3$: a 1% rise in s raises $\bar{k}$ by 1.5% but $\bar{y}$ by only 0.5%. Diminishing returns to capital attenuate the output response relative to the capital response.

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